

EXPERIMENT NO 2

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Batch:- C3

INPUT CODE

```
#include<stdio.h>
void createarray(int arr[],int n)
{
    int i;
    printf("\nEnter %d elements : \n",n);
    for(i=0;i<n;i++)
    {
        printf("Enter element %d: ", i + 1);
        scanf("%d",&arr[i]);
    }
}
void displayarray(int arr[],int n)
{
    int i;
    printf("\n Created Array: ");
    for(i=0;i<n;i++)
    {
        printf("\t%d",arr[i]);
    }
    printf("\n");
}
int linearsearch(int arr[],int n,int key)
{
    int i;
    for(i=0;i<n;i++)
    {
```

```

        if(arr[i]==key)
        {
            return i;
        }
    }
    return -1;
}
int binarysearch(int arr[],int n,int key)
{
    int low=0;
    int high=n-1;
    int mid;
    while(low<=high)
    {
        mid = low + (high - low) / 2;
        if(key==arr[mid])
        {
            return mid;
        }
        else if(key<arr[mid])
        {
            low=0;
            high=mid-1;
        }
    }
    else
    {
        high=n-1;
        low=mid+1;
    }
}
return -1;
}
void displaymenu()
{

```

```

printf("\n\n=====MENU=====");
printf("\n1. Create Array");
printf("\n2. Display Array");
printf("\n3. Linear Search");
printf("\n4. Binary Search");
printf("\n5. Display Menu");
printf("\n6. Exit");
printf("\n=====");
}

```

```

int main()
{
    int arr[100];
    int n=0;
    int choice;
    int key;
    int result;
    displaymenu();
    do
    {
        printf("\nEnter your choice: ");
        scanf("%d",&choice);
        switch(choice)
        {
            case 1 :
                printf("\nEnter the size of an Array : ");
                scanf("%d",&n);
                createarray(arr,n);
                printf("\nArray created successfully");
                break;
            case 2 :
                if(n==0)
                {
                    printf("\nArray has not yet been created.");
                }
            }
        }
    }
}

```

```

    }
    else
    {
        displayarray(arr,n);
    }
    break;
case 3 :
if(n==0)
{
    printf("\nPlease create the array first.");
}
else
{
    printf("\nEnter the value to search:");
    scanf("%d",&key);
    result = linearsearch(arr,n,key);
    if(result!= -1)
    {
        printf("\nValue %d found at position %d.",key,result+1);
    }
    else
    {
        printf("\nValue %d not found in the array.",key);
    }
}
break;
case 4 :
if(n == 0)
{
    printf("\nPlease create the array first.");
}
else
{
    printf("\nNote: Binary Search requires the array to be sorted.");
}

```

```

    printf("\nEnter the value to search:");
    scanf("%d", &key);
    result = binarysearch(arr, n, key);
    if(result != -1)
    {
        printf("\nValue %d found at position %d (Index %d).", key, result + 1,result);
    }
    else
    {
        printf("\nValue %d not found in the array.", key);
    }
}
break;
case 5 :
displaymenu();
break;
case 6 :
printf("\nExiting Program...");
break;
default :
printf("\nInvalid choice.");
}
} while(choice!=6);
return 0;
}

```

OUTPUT

```
Enter the size of an Array : 6

Enter 6 elements :
Enter element 1: 12
Enter element 2: 13
Enter element 3: 14
Enter element 4: 15
Enter element 5: 16
Enter element 6: 17

Array created successfully
Enter your choice: 2

Created Array:      12      13      14      15      16      17

Enter your choice: 3

Enter the value to search:23

Value 23 not found in the array.
Enter your choice: 3

Enter the value to search:12

Value 12 found at position 1.
Enter your choice: 4

Note: Binary Search requires the array to be sorted.
Enter the value to search:23

Value 23 not found in the array.
Enter your choice: 4

Note: Binary Search requires the array to be sorted.
Enter the value to search:14

Value 14 found at position 3 (Index 2).
Enter your choice: 5

=====MENU=====
1. Create Array
2. Display Array
3. Linear Search
4. Binary Search
5. Display Menu
6. Exit
=====

Enter your choice: 6

Exiting Program...

...Program finished with exit code 0
Press ENTER to exit console.
```